Chapter 10. Unsupervised Learning

- Variational Auto Encoder

input x → Encoder → Latent Space → Decoder → output x'

- (i) $z \sim N(0, I_{n \times n})$

  (ii) $x|z \sim N(g(z), \sigma^2 I_{n \times n})$

Note.

Cannot track $p(z|x)$

- $ELBO(x^{(i)}) = \log p(x^{(i)}) - D_{KL}(Q_i \| p_{z^{(i)}|x^{(i)}})$

$$= \mathop{E}_{z \sim Q_i}[\log p(x^{(i)}, z^{(i)})] - \mathop{E}_{z \sim Q_i}[\log Q_i]$$

$$= \sum_{z^{(i)}} Q_i(z^{(i)}) \log \frac{p(x^{(i)}, z^{(i)})}{Q_i(z^{(i)})}$$

Note.

Suppose Q_i Gaussian

(i) $Q_i(z^{(i)})$ ⇒ density of Q_i

(ii) $\mathop{E}_{z^{(i)} \sim Q_i}$ ⇒ sample from Q_i

- $ELBO = \sum_i ELBO(x^{(i)})$

Option 1. $Q_i(z) := Q_i^1(z_1) Q_i^2(z_2) \cdots Q_i^k(z_k)$

with independent coordinate

Option 2. if $z \in \mathbb{R}^k$ is conti latent variable.

assume Q_i to be Gaussian

- $Q_i(z^{(i)}) \approx p(z^{(i)}|x^{(i)})$

  ↳ function of $x^{(i)}$

- for example, let $Q_i = N(\xi_\theta(x^{(i)}), \text{diag}(v_\varphi(x^{(i)}))^2)$

(i) encoder $\xi_\theta(x)$, $v_\varphi(x)$ → optimize ELBO

(ii) decoder $\xi_\theta(z)$

$$\theta := \theta + \eta \nabla_\theta ELBO$$

$$\varphi := \varphi + \eta \nabla_\varphi ELBO$$

$$\phi := \phi + \eta \nabla_\phi ELBO$$

Re-Parameterization

- $z^{(i)} \sim Q_i = N(\xi_\theta(x^{(i)}), \text{diag}(v_\varphi(x^{(i)})^2)$
- $z^{(i)} \sim \xi_\theta(x^{(i)}) + v_\varphi(x^{(i)}) \odot \varepsilon^{(i)}$

with $\varepsilon^{(i)} \sim N(0, I_{n \times n})$

Note.

$\nabla \mathop{E}_{z \sim Q_\theta}[f(\theta)] \neq \mathop{E}_{z \sim Q_\theta}[\nabla f(\theta)]$

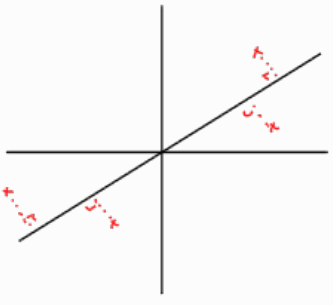
⇒ make θ, φ irrelevant with Q

- dataset $\{x^{(1)}, \dots, x^{(n)}\}$
- x_i와 x_j가 linearly 다르고 서로 같은 값이라면? (상수배)

(i) normalization : $x_j^{(i)} \leftarrow \dfrac{x_j^{(i)} - \mu_j}{\sigma_j}$ with $\mu_j = \frac{1}{n}\sum_i x_j^{(i)}$

$\sigma_j^2 = \frac{1}{n}\sum_i (x_j^{(i)} - \mu_j)^2$

(ii) Principal Components Analysis (PCA)



maximize variance of projected data.

- length of projection $x^T u$ ⇒ 라그랑주 승수법

$\begin{cases} x : \text{point} \\ u : \text{unit vector} \end{cases}$

$L = u^T \Sigma u - \lambda(u^T u - 1)$

$\frac{\partial L}{\partial u} = 2\Sigma u - 2\lambda u = 0$

eigenvector

- $u^T u = 1$ (unit vector)
- variance $= \frac{1}{n}\sum_i (x^{(i)T} u)^2$

$= u^T \left(\frac{1}{n}\sum_i x^{(i)} x^{(i)T}\right) u$

$= u^T \Sigma u$ → covariance

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k개의 eigenvectors

$y^{(i)} = \begin{bmatrix} u_1^T x^{(i)} \\ u_2^T x^{(i)} \\ \vdots \\ u_k^T x^{(i)} \end{bmatrix} \in \mathbb{R}^k$

k차원 approximation.

↓ eigenvectors

i.e k principal components of data

Note. PCA의 활용

- compression : $x^{(i)} \rightarrow y^{(i)}$ 고차원 → 저차원
- preprocess dataset : before supervised learning → avoid overfitting
- Noise Reduction.